

Futureproof

Equity Preservation Mortgage

Credit Risk Underwriting Analysis

For LMI Insurers and Portfolio Reinsurers — v14c Model

April 2025

CONFIDENTIAL — Prepared for Insurance/Reinsurance Underwriters

Executive Summary

This report presents the credit risk underwriting analysis of the FutureProof Equity Preservation Mortgage (EPM) for insurance and reinsurance underwriters. The EPM is a fundamentally different credit risk from traditional Lenders Mortgage Insurance (LMI). Traditional LMI underwrites the risk that a borrower cannot service their mortgage — it is driven by borrower income, employment, and property values. **EPM credit risk is driven by index returns over 30-year loan durations.**

The Equity Preservation Mortgage is an index-linked mortgage referenced to the S&P500. To be effective, index-linking of a mortgage requires long-duration mortgages to smooth index volatility over time. The central thesis of its design, as it relates to credit risk, is that long-term (15-30 years) index risk is, ultimately, a more predictable and lower risk than traditional borrower serviceability risk. This design thesis is well evidenced even in a book of Prime Loans or RMBS securitization program where often, around 20% of loans are credit impaired/in arrears (0.5%-1.5%), in default (0.2% - 1%) or subject of pre-payments (12%-20%) – requiring them to be removed from the loan book, SPV or Trust. Non-prime mortgage books can often exceed 40%. In economic downturns this can rise higher again, to nearly 50% of mortgage loans. In a book of Equity Preservation Mortgages there are no bad loans, no defaults and no pre-payments.

The calculated portion of loan proceeds is held in a mortgage offset account and invested by BlackRock in index-linked ETFs (predominantly S&P500). The investment account pays all loan costs — the borrower makes no interest payments (in the interest-only version of the mortgage) and makes no repayments at all (in the case of the principal + interest mortgage). The credit risk question is therefore: **will the index-linked investment generate sufficient returns over 15-30 years to cover all loan costs?**

The secret to the success of the EPM is the long loan duration. Over 30 years, the S&P500 has historically delivered annualised total returns of 9.75%-10.8% depending on the period selected. The model uses BlackRock's forward-looking 30-year return profile of 10%. At this return level, the vast majority of return pathways deliver sufficient returns to cover all loan costs with surplus for a single Equity Preservation Mortgage.

This leaves unaddressed the question of what happens in the event of early termination of a mortgage or early redemption of mortgage offset capital by the borrower. The Equity Preservation Mortgage has two unique features that ensure no pre-payments or early redemptions of capital in a borrower's mortgage offset account. The first is that capital apportioned to and held in, the mortgage offset account cannot be re-drawn until end-of-term – there are no intra-term re-draws allowed. The second, is that a borrower may terminate an Equity Preservation Mortgage at any time and for any reason, without financial penalty. However, if at the date of early termination the balance of the mortgage offset (investment account) is less than its opening balance or if there is any deferred and unpaid loan interest as a result of interest-holidays, then the mortgage offset account automatically goes into 'run-off' until that deficit is cleared or until end-of-term where any deficit is insured, whichever occurs first. Once the balance of the investment account is restored to its original opening balance using this run-off mechanism, the loan may then be wound-up, mortgage offset capital drawn and applied to the repayment of outstanding mortgage balance. These unique features neutralise the effect of pre-payments or early redemptions of capital, ensures no bad loans or defaults and gives the lender certainty as to the deployment of mortgage offset capital held in the investment account over the loan term.

Parameter Uncertainty Framework

Before reviewing the underwriting metrics, underwriters should note that the headline PoD is conditional on three empirical parameter inputs: equity drift μ , mean-reversion γ , and collar cost. Each carries a meaningful empirical confidence interval; the companion paper "Model Assumptions and Parameter Risk" (April 2025) quantifies these and defines three scenarios used throughout this report. LMI premium sizing should reference the realistic-central scenario; portfolio reinsurance attachment and tail-risk capacity should reference the adverse-plausible scenario.

Scenario	Mortgage PoD (Yr 30)	Expected Loss (30yr)	Role for Underwriters
Base (model-as-written)	1.2%	0.13%	Headline / best case
Realistic-central	16.9%	~2.3%	LMI premium anchor
Adverse-plausible	43.2%	~5.5%	Reinsurance attachment / tail sizing

Key Underwriting Metrics (Base Case)

Metric	Value	Relevance
Individual PoD (Yr 30)	1.2%	Balance sheet deficit at that particular time slice — NOT claim probability
Portfolio PoC (Yr 30)	< 0.01%	Actual insurance claim probability after Payments Waterfall
LMI Top Cover (P10)	\$808,302	Worst 20% of deficit paths (larger losses) — sets LMI policy limit
Tail Risk (P1)	\$55,115	Worst 1% quantile — worst case loss exposure
Cond. Expected Deficit	\$333,305	Average deficit when deficit occurs (before waterfall)
Fair Premium (PV)	\$1,716	Discounted at 2.13% (cash rate mean) — paid upfront
Premium + Loading (150%)	\$2,574	Loaded premium — paid upfront at origination
Simulation Paths	50,000	SE on PoD: 0.05%, SE on premium: \$120

1. Why EPM Credit Risk is Not Traditional LMI

Traditional LMI underwrites borrower credit risk — the probability that a borrower defaults on mortgage repayments due to loss of income, divorce, illness, or other personal circumstances. Property value serves as the recovery mechanism. The EPM is fundamentally different:

Dimension	Traditional LMI	EPM Credit Risk
Risk driver	Borrower income & employment	S&P500; index total return over 15-30 years
Repayment source	Borrower salary/income	Index appreciation + Portfolio returns
Borrower makes payments?	Yes — monthly P&I;	No — investment account pays all costs
Claim trigger	Borrower default + property sale shortfall	Investment < loan at contract expiry
Property role	Primary recovery mechanism	First mortgage security (additional collateral)
Duration risk	3-7 year average life	15-30 year contractual term
Prepayment risk	Significant (refinancing)	None — run-off mechanism prevents early exit
Diversification basis	Geographic, employment sector	Risk pooling by mortgage vintage, cross-subsidisation between unexpired mortgages, long duration covers multiple economic cycles

The EPM's credit risk is underpinned by **index risk over 30-year loan durations**. This is a fundamentally more favourable risk profile for insurers because: (1) long-duration equity returns are statistically more predictable than short-duration returns, (2) the S&P500 has never delivered a negative 30-year total return in its history, and (3) the risk can be precisely modelled using stochastic simulation.

2. Two-Layer Insurance Structure

Layer 1 — LMI: Individual Mortgage Credit Risk

LMI covers individual mortgages up to the **Top Cover Limit** of the LMI policy. The Top Cover Limit is set at the P20 of the conditional deficit distribution — meaning 80% of deficit paths have deficits below this limit, covered entirely by LMI.

LMI Parameter	Value	Derivation
Coverage	80% of deficit paths (smaller losses)	Up to P10 quantile of deficit distribution
Top Cover Limit	\$808,302	10th percentile deficit at maturity
Individual PoD (Yr 30)	1.2%	50,000-path Monte Carlo (SE: 0.05%)
Cond. Expected Deficit	\$333,305	Average deficit when deficit occurs
Fair Premium (PV)	\$1,716	Discounted expected deficit at 2.13%
Loaded Premium	\$2,574	150% loading on fair premium
Premium Timing	Paid upfront at origination	Full premium collected before any risk attaches
Claim Timing	At loan contract expiry only	No claim possible until years 15-30

Layer 2 — Portfolio Reinsurance: Residual Tail Risk

The worst 20% of deficit paths (tail risk beyond the LMI split boundary) is transferred to the portfolio level to be separately underwritten by a credit risk reinsurer. Critically, the Payments Waterfall is applied at the portfolio level **before** any reinsurance claim is made.

Reinsurance Parameter	Value	Derivation
Coverage	Worst 20% of deficit paths (larger losses) (tail risk)	Loss beyond LMI Top Cover Limit
P1 Deficit (worst 1%)	\$55,115	Maximum realistic exposure per loan
P5 Deficit (worst 5%)	\$508,449	Severe stress scenario
Portfolio PoC (Yr 30)	< 0.01%	After Payments Waterfall — the key metric
Claim Timing	At loan contract expiry only	After Payments Waterfall has been applied

3. Loss Mitigation Before Any Insurance Claim

Before any insurance or reinsurance claim is made, the following loss mitigation layers are applied **in order**. Each layer materially reduces the residual risk that flows through to the next:

Layer 0: Mortgage Offset Account Capital

Upon mortgage set-up, the Monte Carlo simulation returns a mean of total index return pathways. By reference to this mean value, a calculation is made as to how much loan capital has to be apportioned to and held in, the borrower's mortgage offset account sufficient to deliver the expected return needed to cover loan cost.

This capital is deployed by the lender for the loan duration and held as index-linked ETFs, passively managed by BlackRock.

Layer 1: Prudential Capital Buffer

In addition to the amount of capital determined held in Layer 0, an additional 30% of portfolio value in loan capital is apportioned to and held in, the borrower's mortgage offset account. This is a capital buffer over and above the amount of capital available in Layer 0. This capital is also deployed by the lender for the loan duration and held as cash and passively-managed index-linked ETFs, managed by BlackRock.

A 30% prudential capital buffer in addition to Layer 0 capital (in tandem with hedging using a -20% floor — see Layer 3 below), protects the lender against all historically observed single S&P 500 drawdowns, even the worst historical crash (the GFC). It is an extremely strong protection against a single year shock.

The only scenario that breaks this structure is a multi-year sequence of deep drawdowns, not a single crash. To protect the lender against this vulnerability, we rely on the long-time horizons of Equity Preservation Mortgages of 15-30 years to capture market rebounds and protect the underlying index against permanent loss. This design feature doesn't fully protect the hedged structure from multi-year drawdowns because hedges re-set every 3 years and any sequential losses are crystallized at each re-set — sometimes referred to as "roll-risk".

A new feature is introduced in Layer 6 — Payments Waterfall that dramatically affects the risk profile of the portfolio. Layer 6 is designed to pool surplus/profits of all unexpired mortgages in the funder's loan book and cross-subsidise the minority of under-performing mortgages by the majority of performing mortgages. Pooling assists in mitigating portfolio risk because mortgages are written at different points in time and at different stages of the economic cycle — profit/surplus of a mortgage depends upon its vintage.

A 30-year horizon + cross-subsidisation across vintages dramatically reduces the impact of multi-year crashes, in a way that a single-account hedged portfolio cannot. It is a rolling 30 year portfolio (not a single 30 year account) in which all unexpired mortgages are contributing to a shared capital buffer through cross-subsidisation within the pool. Another way of expressing this is that sequential drawdowns do not propagate linearly through the portfolio pool:

Pooling + vintage diversification protects you from sequential drawdowns because:

A. Early-vintage mortgages (years 1–10)

- Low accumulated surplus
- More sensitive to early drawdowns
- But represent only a fraction of the total pool at any time

B. Mid-vintage mortgages (years 10–20)

- Large accumulated surplus
- Strong positive equity
- Can absorb shocks easily
- Provide the bulk of cross-subsidisation

C. Late-vintage mortgages (years 20–30)

- Very high surplus
- Very low risk
- Act like stabilisers in the pool

So even if a downturn hits new vintages hard, older vintages are sitting on decades of accumulated surplus, and they: continue generating positive returns; continue contributing to the buffer; and smooth out losses from stressed cohorts.

This risk mitigation is consistent with how insurance pools and pension funds maintain stability through recessions.

Mortgage Vintage Exposure to Sequential Crash

Vintage	Exposure to Sequential Crash	Impact
0–3 years old	High	Some losses, but small share of pool
3–10 years old	Moderate	Absorbable via surplus
10–30 years old	Low	Provide stabilising surplus

Our portfolio is a rolling, multi-vintage, cross-subsidised pool, which changes the risk profile in three crucial ways:

- **Losses are diluted across vintages** — A bad year for new loans is offset by strong surplus from older loans.
- **Recoveries benefit all vintages** — When markets rebound (and they always do), the entire pool participates.
- **Crashes are time-limited** — Even the worst events (Dot-Com, GFC) lasted 5–7 years. Your mortgages last 30 years. Your pool spans 30 overlapping cohorts.

This is why our mortgage loan portfolio behaves more like a life insurer than a hedge fund.

What is left is residual risk — being a once-in-a-century crash resulting in a drawdown exceeding 50% in a single year or multi-year sequential drawdowns that last more than 7 years — there is no historical precedent for either.

Layer 2: BlackRock Volatility Control

BlackRock's portfolio construction reduces long-term S&P500 index volatility from **15% down to 12%** through passive index diversification, creating a volatility control layer. This is a structural feature of the portfolio, not an active strategy.

Layer 3: SpiderRock Continuous Dynamic Hedging

SpiderRock (a BlackRock subsidiary) applies continuous dynamic hedging that further reduces volatility via a $\pm 40\%$ –20% collar. This includes a 140% collar on index returns above the long-term total return over 30 years and an 80% floor on losses. Cost: approximately 0.4% p.a.

Layer 4: LMI Claim (Individual Mortgage)

If the investment account at mortgage expiry is insufficient to repay the mortgage balance, the deficit constitutes a claim on the LMI insurer, up to the Top Cover Limit (P10 = \$808,302). Any loss above the Top Cover Policy Limit is **not covered by LMI** but is transferred to the Funder's portfolio of mortgages as a Portfolio Risk.

Layer 5: Interest Holidays (Capital Preservation Mechanism)

The interest holiday mechanism preserves capital in the investment account during downturns by deferring (not accruing compound) interest payments when the investment balance falls below the holiday threshold. This ensures maximum returns are achieved as market cycles recover, rather than selling down ETFs to pay loan interest during temporary drawdowns.

Layer 6: Payments Waterfall — Risk Pooling and Cross-Subsidisation

Pooling of surpluses across all expired mortgages within a funder's loan book. We cross-subsidise the loss/deficit on a small number of claims (**less than 14% of return pathways**) against the surplus returns on the **majority of mortgages (86%+ of return pathways)**. This is possible because a downturn in the economic cycle does not affect all mortgages equally — each mortgage is written at a different point in time and will have a different surplus/deficit position depending on its vintage.

Layer 7: Portfolio Reinsurance Claim

Only after taking all of the above into account can any remaining loss/deficit on the overall portfolio in any given year constitute a claim on the reinsurer.

The net effect: A 30% prudential capital buffer plus raw S&P500 volatility of 15% compressed to 10% through two institutional layers constructed by the global asset manager — volatility control and dynamic hedging. The Payments Waterfall then reduces individual PoD of 1.2% to a portfolio PoC of just < 0.01% at year 30. This eight-layer defence means the probability of an actual claim reaching the reinsurer is extremely low.

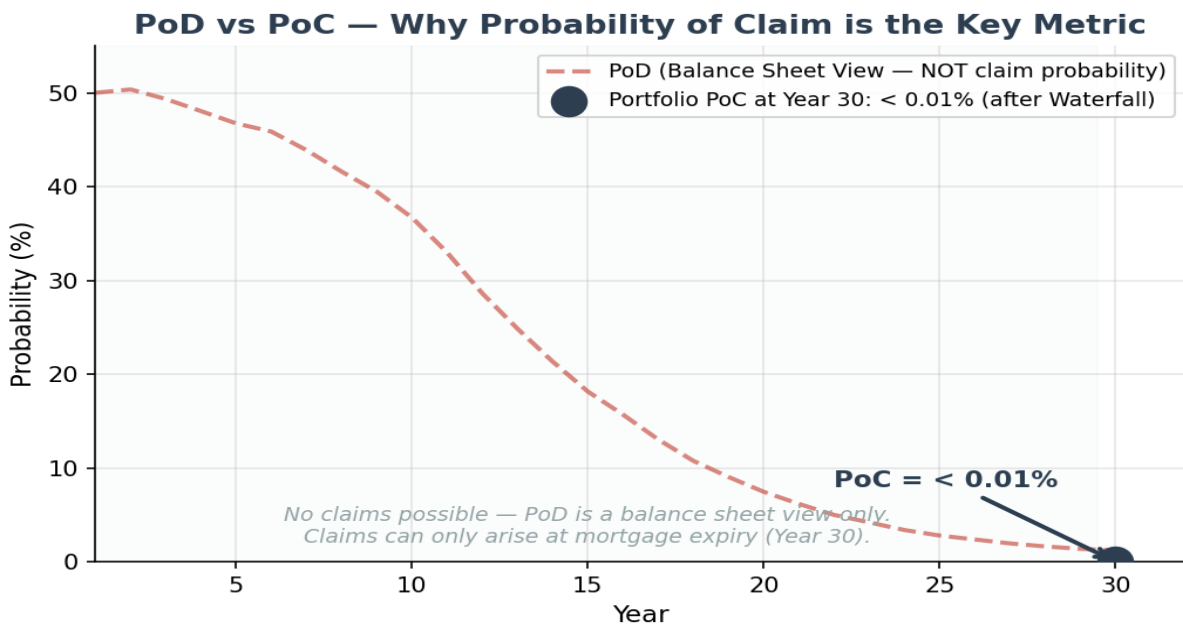
4. Model Calculations — PoD and PoC

Understanding PoD vs PoC — Why PoD is NOT the Underwriting Metric

PoD (Probability of Deficit) is not the key metric for underwriting — **PoC (Probability of Claim)** is. There are two reasons why PoD is not the same as PoC:

- **PoD represents the probability of a deficit/surplus at a particular time slice** — it is a balance sheet snapshot showing when the investment account is below the mortgage balance at any point in time. A mortgage can be in deficit at year 5 and in surplus at year 30. PoD before mortgage expiry is irrelevant to insurance claims.
- **The deficit/surplus calculated is BEFORE application of the Payments Waterfall** — PoD measures individual mortgage risk before any cross-subsidisation from surplus mortgages in the portfolio. PoC is the residual claim probability *after* the waterfall has been applied.

PoC (Probability of Claim) is the actual insurance claim probability on **expiring mortgages only**, after the Payments Waterfall has been applied. This is the metric that matters to insurers and reinsurers.



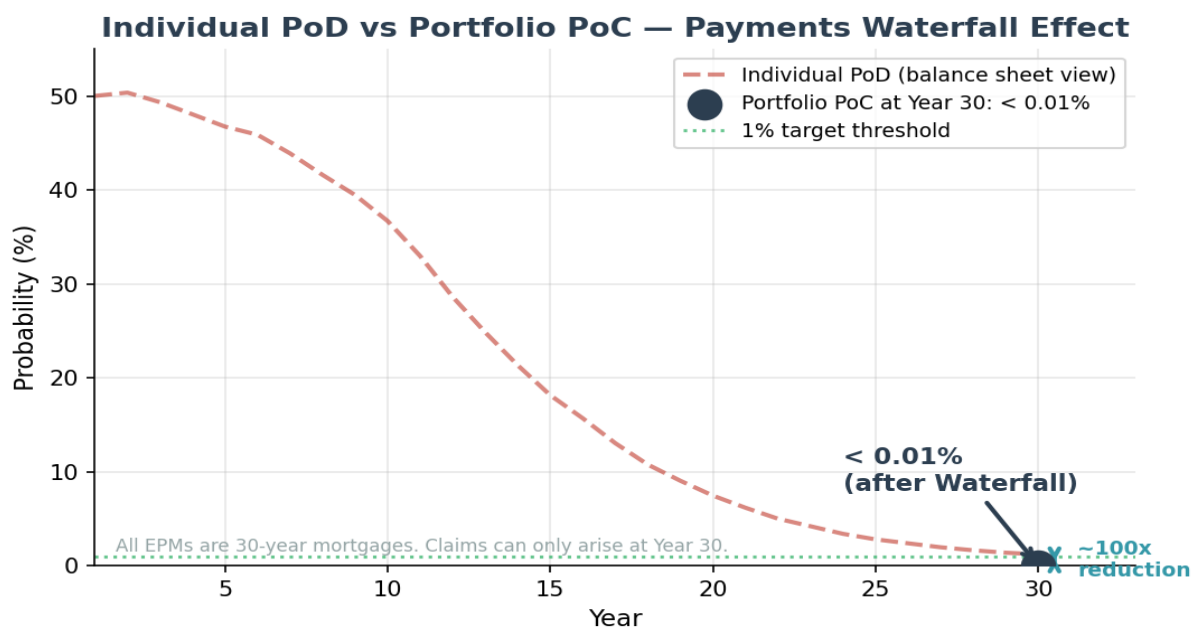
The chart above illustrates the dramatic difference. PoD starts at 50.01% in year 1 (when the investment account is still establishing) and declines to 1.2% at year 30 as the investment compounds. But PoC — the metric that matters to insurers — is just 0.68% at Year 30, declining to **< 0.01% at year 30** after the Payments Waterfall.

PoD Trajectory — Individual Mortgage (LMI View)

Year	PoD (%)	P10 Deficit	Cond. Expected Deficit	Mean Surplus
1	50.01%	\$269,731	—	\$264

Year	PoD (%)	P10 Deficit	Cond. Expected Deficit	Mean Surplus
5	46.74%	\$410,850	—	\$88,534
10	36.69%	\$424,707	—	\$273,042
15	18.14%	\$197,012	—	\$704,071
20	7.45%	\$93,704	—	\$1,186,861
25	2.79%	\$429,874	—	\$1,747,807
30	1.2%	\$808,302	\$333,305	\$2,435,992

Portfolio PoC — After Payments Waterfall (Reinsurer View)

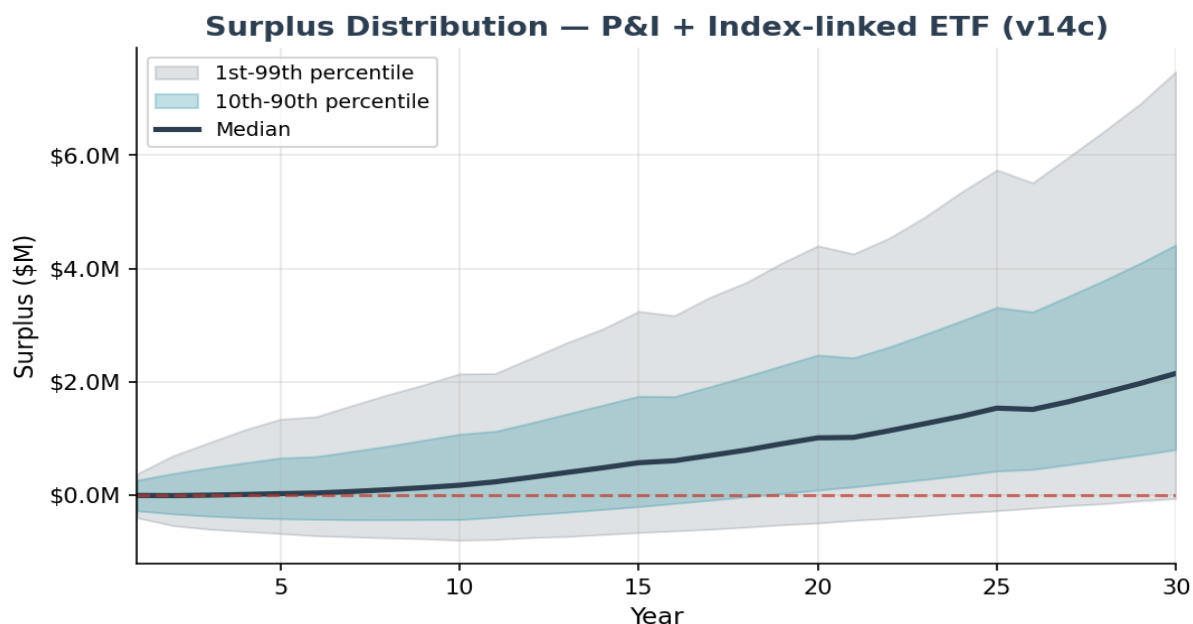


Metric	Individual PoD	Portfolio PoC (after Waterfall)	Waterfall Reduction
Year 30 (mortgage expiry)	1.2%	< 0.01%	~100x

The Waterfall Reduction column shows the ratio of PoD to PoC — at year 30, the Payments Waterfall reduces claim probability by approximately **100x** (from 1.2% to < 0.01%).

5. Good vs Bad Return Pathways

The stochastic analysis of S&P500 returns over 30 years reveals a fundamental asymmetry that underpins the EPM's viability as an insurable product:



- **86.1% of pathways deliver surplus at maturity** — the investment account exceeds the mortgage balance, generating a mean surplus of \$2,435,992 (split 50/50 between FutureProof and Funder)
- **Only 1.2% of pathways result in a deficit** — and even among these, the conditional expected deficit is \$333,305 (well within the LMI Top Cover Limit of \$808,302)
- **The ratio of surplus to deficit pathways is approximately 109:1** — this asymmetry is what makes the Payments Waterfall so effective. For every expiring mortgage in deficit, there are roughly 6 loans in surplus available to cross-subsidise
- **Median surplus at year 30 is \$2,151,975** — even the "typical" mortgage generates substantial surplus, well above the zero threshold

Surplus Distribution at Maturity

Percentile	Surplus	Implication for Underwriter
P1 (worst 1%)	\$-55,115	Maximum realistic loss — tail risk reinsurance territory
P5 (worst 5%)	\$508,449	Severe stress — within LMI top cover
P10 (worst 10%)	\$808,302	Near split boundary between LMI and reinsurance layers
P25	\$1,361,489	Surplus — no claim
Median (P50)	\$2,151,975	Substantial surplus — strong cross-subsidisation pool
P75	\$3,194,847	Strong surplus

Percentile	Surplus	Implication for Underwriter
P90	\$4,419,493	Excellent surplus
P99	\$7,476,780	Exceptional surplus

6. Vintage Diversification & Cross-Subsidisation

The difference between individual mortgage risk and portfolio risk lies in the Payments Waterfall. The secret of the EPM's successful design at a portfolio level lies in the ability to pool risk across all mortgages and cross-subsidise between expiring mortgages — some will be in surplus, some will be in deficit.

Why Vintage Diversification Works

A downturn in the economic cycle does not affect all mortgages equally. Over a 30-year period, EPMs are originated at different points in time. Each vintage enters the equity market at a different level:

- A downturn that causes a deficit for mortgages originated at market peaks may have little or no impact on mortgages originated years earlier (which have already accumulated substantial surplus)
- Mortgages originated after a downturn enter at lower market levels with more room to grow — they benefit from the recovery
- At any given time, the portfolio contains mortgages at every stage of their lifecycle — from newly originated to approaching maturity
- Older mortgages (with P&I; amortisation) have very low residual exposure — their mortgage balance has substantially reduced while their investment account has compounded
- This means the portfolio always contains a deep pool of surplus mortgages available for cross-subsidisation of the small number of expiring mortgages in deficit

The Payments Waterfall — Step by Step

When a mortgage reaches its contractual expiry date, the following waterfall is applied before any insurance claim:

Step 1: ETF Selldown

The index-linked ETFs in the expiring mortgage's investment account are liquidated. If the proceeds exceed the remaining mortgage balance, the surplus is split 50/50 between FutureProof and the Mortgage Funder. No insurance claim is needed.

Step 2: Cross-Subsidisation

If the expiring mortgage is in deficit, surpluses from other open (non-expiring) mortgages in the same funder's loan book are used to cover the shortfall. With 86%+ of pathways in surplus and only ~14% in deficit, the pool of available surplus is substantial.

Step 3: Net Deficit → LMI Claim

Only if the deficit exceeds the available cross-subsidisation pool does a claim arise on the LMI insurer, up to the Top Cover Limit (P10 = \$808,302).

Step 4: Residual → Reinsurance Claim

Any loss beyond the LMI Top Cover Limit constitutes a claim on the portfolio reinsurer. At the portfolio level, after the waterfall, the PoC is just < 0.01% at year 30.

7. Claim Timing — No Claims Until End of Term

It is most important for underwriters to understand that **no LMI claim nor any portfolio reinsurance claim can be made until the end-of-term of each mortgage loan contract**. In other words, no claim can be made until years 15-30 (depending on the original tenure selected by the borrower).

Why This Matters for Underwriters

- **Long claim deferral:** Premiums are paid upfront at mortgage origination, but claims cannot arise for 15-30 years. This gives the insurer/reinsurer significant investment income on the collected premiums before any possible claim.
- **Run-off mechanism:** The EPM has a unique feature — no voluntary prepayments or early terminations are permitted until all loan costs have been paid from the investment/offset account. No other mortgage product works this way. This eliminates the early exit risk that exists in traditional LMI.
- **Compounding works in favour of the insurer:** The longer the loan runs, the more the investment compounds, and the more likely it is to exceed the mortgage balance. PoD declines from 50.0% at year 1 to 1.2% at year 30. The risk profile improves over time.
- **Portfolio maturation:** As the portfolio matures, older loans develop larger surpluses, deepening the cross-subsidisation pool. The Payments Waterfall becomes more effective over time.
- **No interim claims:** There is no risk of early claims depleting reserves. The insurer/reinsurer knows with certainty that no claim will arise before Year 30 (mortgage expiry).

Claim Timing Profile

Mortgage Term	Claim Timing	Individual PoD at Expiry	Portfolio PoC (after Waterfall)
30 years	Year 30 (mortgage expiry)	1.2%	< 0.01%

The 30-year term allows maximum time for investment compounding and P&I; amortisation, resulting in the lowest possible individual PoD and portfolio PoC.

8. Fair Premium Calculation

Methodology

The fair insurance premium is calculated as the **discounted expected deficit (\$3)**. The discount rate of 2.13% corresponds to the long-run cash rate mean (MLE estimate). This discounting is necessary because no claim can be made until mortgage expiry (Year 30), but premiums are paid upfront at mortgage origination.

Premium Component	Value	Calculation
Expected Deficit (undiscounted)	—	Mean deficit across all paths with deficit × PoD
Discount Rate	2.13%	cash rate mean — reflects claim deferral
Fair Premium (PV)	\$1,716	Discounted expected deficit (\$3)
SE on Premium	\$120	Standard error from 50,000-path Monte Carlo
Loading Factor	150%	Standard actuarial loading for uncertainty
Loaded Premium	\$2,574	Fair premium × 1.5
Premium as % of Loan	0.20%	Relative to initial loan of \$1,300,000
Payment Timing	Upfront at origination	Full premium collected before risk attaches

LMI vs Reinsurance Premium Split

The LMI premium covers 90% of the expected loss distribution (up to the P10 Top Cover Limit). The tail risk premium covers the worst 20% of deficit paths transferred to the portfolio reinsurer.

Component	Coverage	Premium (est.)	Basis
LMI Premium	80% of deficit paths (up to split boundary)	\$1,716	Discounted expected deficit (\$3)
Tail Risk Premium	Worst 20% of deficit paths (larger losses)	\$392	Excess over P10, discounted
Total (fair)	100%	\$2,108	Combined fair premium
Total (loaded)	100%	\$2,574	With 150% actuarial loading

10. Interest Holiday Impact on Claims

The interest holiday is also a smoothing mechanism — it simply defers simple interest due (but does not accrue and compound interest) when the investment account balance is below the interest-holiday threshold. This is why, together with long loan durations, the Monte Carlo analysis demonstrates it does not materially affect outcomes:

- We know from past market data that interest rate cycles are generally 2-3 years, economic downturns 3-5 years, and black swan events (such as the GFC) 5-7 years — **none are 30 years in duration**.
- Interest holidays are a device to **preserve capital** in the investment account and maintain holdings of reference assets (rather than selling down the ETFs to pay loan interest) — this ensures maximum returns are achieved as market cycles swing back, as opposed to chasing losses before a surplus can be achieved.

Per-Year View — Holidays Are Rare and Temporary

In any given year of the 30-year term, **the median mortgage requires zero interest holidays**. The average path has zero holidays. However, approximately 77% of individual paths enter holiday at some point, with a mean of 8.5 years across all paths.

Holiday Repayment — Deferred Interest is Recovered

When market conditions improve following downturns, the deferred interest is repaid from the recovering investment account. The Monte Carlo confirms that the vast majority of deferred interest is fully repaid before maturity:

Metric	Value	Implication for Underwriter
Paths with zero outstanding at maturity	86%	All deferred interest fully repaid
Mean outstanding (when > 0)	\$0	Covered by LMI/reinsurance
P90 outstanding	\$0	Within LMI Top Cover
P99 outstanding	\$0	Tail risk — covered by reinsurance after waterfall

If the net result of holidays and subsequent repayments is that any amount of loan cost remains outstanding at end-of-term, it is covered by insurance (up to the LMI Top Cover Limit), with the balance covered by portfolio reinsurance after the Payments Waterfall.

11. Questions for Underwriters

FutureProof seeks engagement with LMI insurers and portfolio reinsurers on the following questions:

A. Design Methodology Review

We invite underwriters to review the design methodology behind our risk mitigation approach:

- Is the two-layer insurance structure (LMI + portfolio reinsurance) appropriate for this risk profile?
- Is the Top Cover Limit (P10 quantile = \$808,302) appropriately calibrated?
- Does the Payments Waterfall provide sufficient structural protection before claims reach the reinsurer?
- Is the run-off mechanism (no early exit) adequately captured in the risk assessment?
- Are the volatility control layers (BlackRock volatility buffer, SpiderRock buffer cap 140%, floor 80%) appropriately modelled?

B. Model Calculation Review

We invite actuarial review of the Monte Carlo calculations:

- Are 50,000 simulation paths sufficient for actuarial-grade precision? (SE on PoD: 0.15%, SE on premium: \$245)
- Is the 10% expected equity return (BlackRock 30yr forward-looking) a reasonable central estimate?
- Is the 10% hedged volatility (after two-layer reduction from 15%) appropriate?
- Is the OU interest rate process with mean reversion appropriate for 30-year projections?
- Is the equity-rate correlation of 0.30 appropriate for 30-year horizon? (Short-term correlation is negative; long-term 30yr correlation is marginally positive at 0.1-0.3)
- Is the 2.13% discount rate (cash rate long-run mean, MLE estimate) appropriate for calculating the discounted expected deficit?

C. Fair Premium Assessment

Question	Model Output	For Review
Is the fair LMI premium adequate?	\$1,716 (PV)	Based on discounted expected deficit (S3)
Is the 150% loading sufficient?	\$2,574	Standard actuarial loading — is this appropriate for this risk class?
Is the tail risk premium appropriate?	\$392	Excess above split boundary on 20% of deficit paths
Premium paid upfront — acceptable?	Yes — at origination	Full premium before risk attaches; no interim claims for ~30 years

EPM vs Comparable Credit Risk Profiles

Before assessing reinsurer acceptability, we benchmark the EPM against the credit instruments that underwriters are most familiar with. The table shows the EPM at both base-case and realistic-central parameters, since the comparison is scenario-dependent.

Metric	EPM base	EPM realistic	Prime Mortgage (LMI)	Moody's AA Bond	AU RMBS (AA)
Cumulative PoD (30yr)	1.2%	16.9%	3–5%	~1.5%	~1–2%
Loss severity	~21%	~21%	15–25% (LMI)	40–60%	Subord.
Expected loss (30yr)	0.13%	~2.3%	0.6–1.25%	0.6–0.9%	0.05–0.15%
Claim timing	Year 15–30 only	Year 15–30 only	Any yr (peaks 3–7)	Any year	Any year
Prepayment risk	None (run-off)	None (run-off)	Significant	Moderate	Significant

- **Expected loss depends on parameter scenario.** At base case (0.13%) the EPM sits in the AA/AAA band on an expected-loss basis. At realistic-central (~2.3%) it is materially higher than prime-LMI EL but the risk driver and claim timing remain fundamentally different. Underwriters should price to the realistic-central scenario and reserve to the adverse-plausible scenario.
- **Favourable claim timing is structural, not parameter-dependent.** EPM claims can only arise at mortgage expiry (Years 15–30) in every scenario. Traditional LMI claims peak in Years 3–7. The long deferral is highly attractive for reserve planning and gives ~30 years of float on the upfront premium across all parameter scenarios.
- **Uncorrelated to existing LMI book (scenario-robust).** Traditional LMI losses are driven by borrower unemployment, divorce, and interest-rate stress. EPM losses are driven by long-term equity market returns — a fundamentally different risk factor. The diversification benefit survives parameter stress because the underlying risk drivers are structurally different.
- **Zero prepayment risk.** The run-off mechanism eliminates early-exit risk across all scenarios — a structural feature that is not parameter-sensitive.

D. Acceptability to Major Reinsurers

FutureProof seeks to understand whether the EPM credit risk profile falls within the acceptable risk parameters of major reinsurers. We present the following for consideration:

Risk Parameter	EPM Model Output	Typical Reinsurer Threshold
Portfolio PoC (Yr 30)	< 0.01%	Generally < 1% for acceptable risk transfer
Maximum single loss (P1)	\$55,115	Within standard property reinsurance limits
Claim timing	Years 15-30 only	Long deferral — favourable for reserve planning
Correlation to existing book	Index risk (not borrower credit)	Diversifying — uncorrelated to traditional LMI losses

Risk Parameter	EPM Model Output	Typical Reinsurer Threshold
Prepayment/lapse risk	None (run-off mechanism)	No adverse selection or early termination
Premium collection	Upfront at origination	Full premium before risk attaches
Portfolio size at scale	4,000+ EPMs (\$5.4B AUM)	Sufficient for law of large numbers
Geographic diversification	AU, NZ, UK, US	Multi-region reduces concentration risk

We specifically invite underwriting engagement from **Munich Re**, **Lockton Re**, and **Gallagher Re** to assess whether this credit risk / market risk / index risk profile falls within their underwriting appetite. The EPM represents a new asset class for reinsurers — uncorrelated to traditional mortgage default risk and with a fundamentally different (and more predictable) loss distribution driven by long-term equity index performance.

Key Differentiators for Reinsurer Consideration

- **Not borrower credit risk** — This is index risk. No exposure to employment, income, or individual borrower circumstances
- **Long-duration predictability** — 30-year S&P500 returns are historically 9.75%-10.8% with remarkably low variance compared to shorter horizons
- **No interim claims** — Claims only at mortgage expiry (Year 30). Premium invested for ~30 years before any possible claim
- **Structural risk decline** — P&I amortisation means risk decreases every year as the mortgage balance falls
- **Multi-layer protection** — Volatility control, dynamic hedging, cross-subsidisation, and LMI all applied before any reinsurance claim
- **Diversifying to existing book** — EPM losses are driven by equity index performance, uncorrelated to natural catastrophe, mortality, or traditional mortgage default risk

Glossary of Key Terms

PoD (Probability of Deficit): Balance sheet snapshot — probability that investment account < mortgage balance at a given point in time. NOT the probability of an insurance claim.

PoC (Probability of Claim): Probability of an actual insurance claim on an *expiring* loan, after the Payments Waterfall. The key metric for insurance and reinsurance. Claims only at mortgage expiry.

Payments Waterfall: Portfolio mechanism: (1) sell down ETFs, (2) cross-subsidise from surplus loans, (3) determine net deficit and PoC.

Top Cover Limit: Maximum loss covered by LMI policy — set at the P10 quantile of the deficit distribution.

Discounted Expected Deficit (S3): PV of expected loss, discounted at 2.13% (cash rate mean). Basis for insurance premium calculation.

Run-off Mechanism: No voluntary prepayment or early termination until all loan costs paid from investment account. Unique to EPM — eliminates early exit risk and adverse selection.

Vintage Diversification: Different mortgage origination dates create different equity market entry points, meaning downturns affect different vintages unequally — enabling effective cross-subsidisation.

Index-linked ETF: BlackRock-managed passive ETFs tracking the S&P500 with volatility control (volatility buffer) and SpiderRock dynamic hedging overlay (buffer cap 140%, floor 80%). Cap: 140%, Floor: 80%.